



EDU TERIA

72nd BPSC

Prelims Online Batch

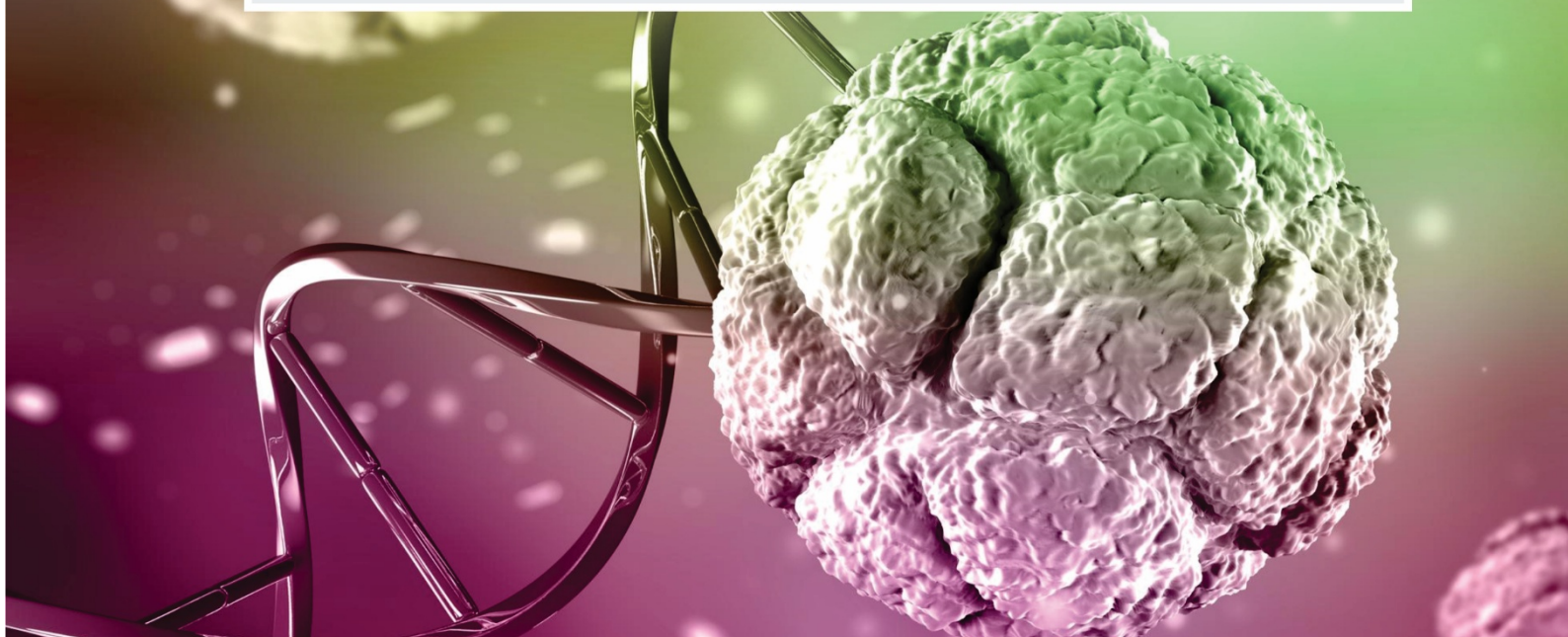
BIOLOGY

Topic wise
Notes

LECTURE

26

**Infectious and
non-Infectious Diseases**



Infectious and non-Infectious Diseases

Human Health & Diseases

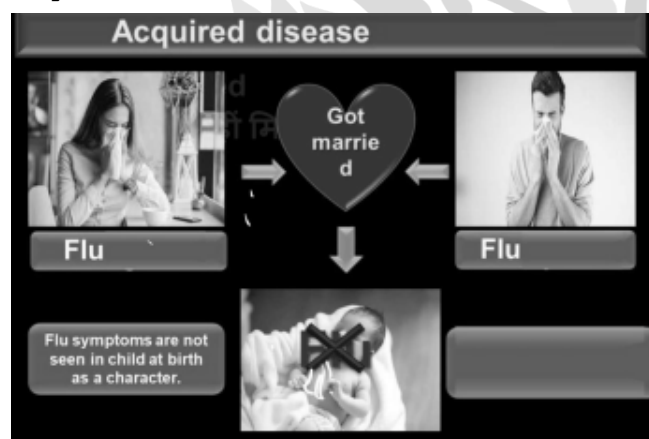
According to WHO: Health is a state of complete physical, mental and social well-being, and not merely an absence of disease or infirmity.

Any change from the normal state that causes discomfort or disability or impairs health is called disease

Type of diseases

Diseases	
Congenital Or Genetic disease	Acquired disease

Acquired disease:



- Not present at birth (Non congenital)
- Developing after birth in response of entry of any pathogens (bacteria, virus, protozoa etc.) inside body. **Ex-** Influenza
- Develops in response of any vital nutrient deficiency in diet. **Ex-** Vitamin A deficiency causes night blindness.
- Developments with ageing
- **Ex-** High blood pressure, loss of memory.
- Develops due any internal abnormality of body like hormonal imbalance, abnormal growth of cell. **Ex-** Cancer, Goitre

Type of Acquired disease

Acquired disease	
Communicable or infectious disease	Non-communicable or noninfectious disease

Infectious disease:

- Diseases which are easily transmitted from one person to another are called infectious diseases.

Bacterial disease:

- Pathogen is bacteria.

Pathogen based disease.

Example:

- Tuberculosis
- Diphtheria
- Pertussis
- Cholera
- Pneumonia
- Tetanus

Bacteria based :

Disease	Causative Agent
Tuberculosis	Mycobacterium tuberculosis
Cholera	Vibrio cholerae
Typhoid	Salmonella typhi
Diphtheria	Corynebacterium diphtheriae
Tetanus	Clostridium tetani
Plague	Yersinia pestis

Proto

Example:

- Typhoid
- Leprosy
- Plague

Tuberculosis

❑ Pathogen:

- Mycobacterium tuberculosis

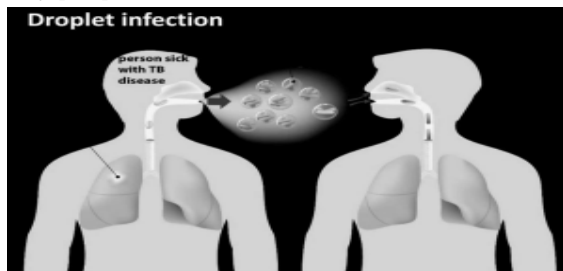
❑ Mode of infection

- Airborne
- Droplet infection



**Droplet infection**

Droplet infection occurs when bacteria or viruses are released into the air through sneezing, coughing, or speaking, landing on the mucous membranes of nearby people.



❖Symptoms :-



Chronic cough for more than 1 week

❖Symptoms:-



Fever

❖Symptoms:-



Weakness

❖Symptoms:-



Bloody sputum

❖Symptoms:-



Weight loss is specific symptoms

❖Primary affecting organ :-



Lungs

❖Test for diagnosis:-



Mounstex test

Tuberculosis:

❑ Treatment:-

- Direct observation Treatment short course (DOTS).
- Combination of many antibiotics is given in direct observation of any family member or volunteer.

Diphtheria:

❑ Pathogen :-

- *Corynebacterium diphtheriae*

❑ Mode of infection

- Airborne
- Droplet infection

Droplet infection



❖Symptoms



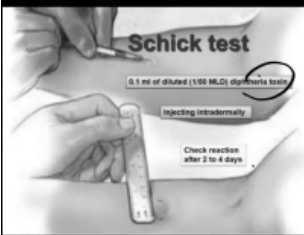
Shore throat,
Difficulty in breathing

❖Symptoms



High grade fever

❖Test for diagnosis:-



Schick test

Treatment :-

Treatments include antibiotics and an antitoxin that neutralizes the diphtheria toxin.

Pneumonia

Pathogen: Streptococcus pneumoniae

Mode of infection

- Airborne Droplet
- infectionDroplet infection:

❖Symptoms :-



Difficulty in breathing

❖Symptoms:-



High fever

❖Symptoms:-



High fever

❖Symptoms:-



Cough

❖Symptoms:-



Headache

❖Symptoms:-



In severe case blue nail and lips

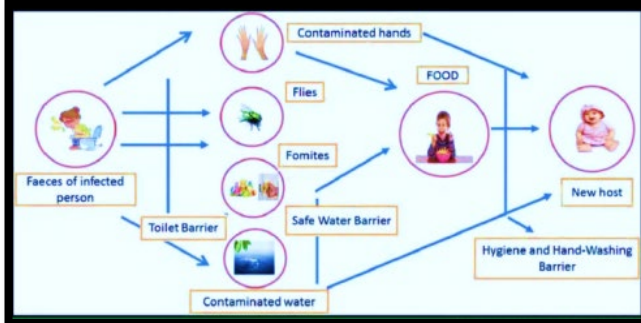
**Pneumonia****Test for diagnosis:**

- Blood test(TLC)
- Chest X-Ray/
- Treatments include antibiotics.



**Cholera:****Pathogen:** *Vibrio cholerae***Mode of infection**

- Spread through contaminated food and contaminated water (waterborne).
- Faecal - oral route transmission

**❖Symptoms:-****Vomiting****❖Symptoms:-**

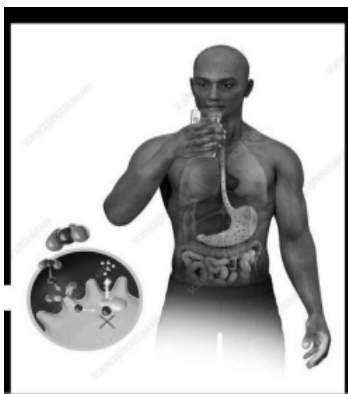
HUMAN BODY IS 70% WATER

24% BONE	75% BRAIN
80% SKIN	90% LUNGS
85% BLOOD	75% MUSCLE



SYMPTOMS OF DEHYDRATION

DIZZINESS	DRY MOUTH
DARK URINE	HEADACHE
THIRST	FATIGUE

**Dehydration****Intestinal tract****Test for diagnosis:**

- Stool test to confirm presence of bacteria
- Rapid cholera "dipstick tests"

❖Treatment :-**Treatment:-**

- Intravenous fluids in severely dehydrated people.
- Antibiotics are also recommended sometimes.

Vaccine :-

- Dukoral, Shanchol, Euvichol

Typhoid**Pathogen :** *Salmonella typhi***Mode of infection**

- Spread through contaminated food and contaminated water (waterborne)

❖Symptoms:-**Sustained high Fever (103–104°F)****❖Symptoms:-****Stomach pain and constipation**



❖Symptoms:-



Loss of appetite



Vomiting

❖Symptoms:-

Headache
& Weakness

Affecting Organ:

First infection in Intestine

- Liver
- Spleen
- Gall bladder
- Kidney

❖Test for diagnosis:-



Widal Test (Slide Agglutination)

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Widal test

Treatment:

- Treatments include antibiotics.

Vaccine:

- TCV (Typhoid Conjugate Vaccine)
- Indian brand Typebar TCV

Fungal based disease

Disease	Causative Agent
Ascariasis	Ascaris lumbricoides
Filariasis	Wuchereria bancrofti
Taeniasis	Taenia (tapeworm)
Hookworm disease	Ancylostoma

